MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) Regular Examination — March 2026

Fourth Semester

Course Code MG4DSCPHY200

Question Paper Code 25901435

WAVE OPTICS

Scheme for Valuation with Model Answers

Reference: Optics *by Eugene Hecht & A.R.Ganesan (5th Edition)*

Mark Distribution Summary

Maximum Marks: **50** Duration: **1 hour 30 minutes**

Part	Question Type	No. to Attempt	Marks Each	Total
Part A	Short Answer Questions	Any 10 out of 12	2	**20**
Part B	Short Essay	Any 6 out of 8	5	**30**
			TOTAL MARKS	**50**

PART A Short Answer Questions [2 × 10 = 20 Marks]

Answer any 10 questions. Each question carries 2 marks.

Q1

What is the relation between wave vector and wavelength? [U / CO1]

The wave vector **k** is a vector quantity that points in the direction of propagation of the wave and whose magnitude **k** (the wave number) encodes the spatial frequency of the wave. Its magnitude is related to the wavelength λ by (**Hecht, §2.2**):

$$k = \frac{2\pi}{\lambda}$$

where k is in $\mathbf{rad\,m^{-1}}$ and λ is the wavelength in **meters**. As a vector, $\mathbf{k} = \frac{2\pi}{\lambda}\,\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit vector along the propagation direction. Larger **k** implies shorter λ (more oscillations per unit length).

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct relation $\mathbf{k} = 2\pi/\lambda$ with units / symbols	2	**2**

Q2

Formulate the wave equation in three dimensions. [K / CO1]

The three-dimensional wave equation governs the propagation of electromagnetic waves and is obtained directly from Maxwell's equations in a linear, isotropic, homogeneous medium (**Hecht, §2.8**). For any scalar component ψ of the electric or magnetic field:

$$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

Expanding the Laplacian ∇^2 in Cartesian coordinates:

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

Here $v = c/n$ is the phase velocity or speed of the wave in the medium of refractive index n ($v = c$ in vacuum). A general plane-wave solution is $\psi(\mathbf{r}, t) = A\exp\left[i(\mathbf{k}\cdot\mathbf{r} - \omega t + \phi_0)\right]$.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct 3D wave equation with Laplacian notation or in Cartesian expansion	1	**2**
v is the phase velocity or speed of the wave in the medium	1	

Q3

Mention any four quantities that describe a wave. [K / CO1]

A sinusoidal wave is completely characterised by the following measurable parameters (**Hecht, §2.2**):

- **Amplitude** (A) — Strength of the electric (or magnetic) field. Determines the intensity (brightness) of light: $I \propto A^2$.
- **Wavelength** (λ) — The distance between two successive points having the same phase in a wave (e.g., crest to crest or trough to trough).; $\lambda = v/\nu$.
- **Frequency** (f) — Number of complete oscillations per second (Hz); $\nu = 1/T$.
- **Phase velocity** (v) — Velocity or speed of a phase front: $v = \nu\lambda = \omega/k$.
- **Phase** (ϕ) — Initial phase angle determining the wave's starting point.
- **Wave number** (k) — Spatial angular frequency.($k = 2\pi/\lambda$) It represents the number of radians per unit length.

(Any four of the above are accepted.)

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any four correct quantities named	1	**2**
Brief description or equation for each quantity	1	

Q4

In a polarising device, what is referred to as the extinction axis? [U / CO2]

A linear polariser (e.g., a Polaroid sheet) has two orthogonal characteristic directions (**Hecht, §8.2.1**). The **extinction axis** (also called the *absorption axis*) is the direction **perpendicular to the transmission axis**, along which the electric field component is completely absorbed by the polariser. Light whose polarisation is aligned with the extinction axis is fully blocked.

This is quantified by **Malus's Law**: when polarised light of irradiance I_0 passes through a polariser at angle θ to the transmission axis,

$$I = I_0 \cos^2\theta$$

At $\theta = 90°$, the incident polarisation is aligned with the extinction axis and $I = 0$ (complete extinction).

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Extinction axis = direction perpendicular to transmission axis; absorbed component	2	**2**

Q5

Why is quartz said to be optically active? [U / CO2]

Quartz(SiO_2) is said to be optically active because it rotates the plane of polarization of linearly polarized light passing through it. This occurs due to different propagation speeds of right- and left-circularly polarized components arising from its asymmetric crystal structure.

Quartz is optically active because its crystal lattice possesses a **helical (chiral) structure** with no plane of symmetry (**Hecht, §8.7.1**). This chirality causes the refractive indices for left- and right-circularly polarised (LCP and RCP) components to differ: $n_{\mathscr{L}} \neq n_{\mathscr{R}}$.

Since any linearly polarised wave is a superposition of LCP and RCP waves, these two components travel at different speeds through the crystal. After propagating a thickness d, they accumulate a phase difference, which manifests as a **rotation of the plane of polarisation**:

$$\beta = \frac{\pi\, d}{\lambda_0}(n_{\mathscr{L}} - n_{\mathscr{R}})$$

There are two mirror-image crystal forms of quartz: *dextrorotatory* ($+ve$, rotates right) and *laevorotatory* ($-ve$, rotates left).

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Definitions for optically active crystal	1	**2**
Rotation formula and optical activity explained	1	

Q6

What phase difference corresponds to a path difference of $\lambda/2$? [U / CO3]

The relationship between optical path difference Δ and the corresponding phase difference δr follows from the fact that one wavelength λ corresponds to one full oscillation cycle, i.e., a phase change of 2π radians (**Hecht, §9.1**):

$$\delta = \frac{2\pi}{\lambda}\,\Delta r$$

Substituting $\Delta r = \lambda/2$:

$$\delta r = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi \text{ rad} = 180^\circ$$

A phase difference of π causes two equal-amplitude waves to **interfere destructively** (complete cancellation).

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct formula $\delta = (2\pi/\lambda)\Delta$	2	**2**

Q7

Explain the formation of Newton's rings. [K / CO3]

Newton's rings arise from **thin-film interference** in a wedge-shaped air gap formed between a plano-convex lens resting on a flat glass plate (**Hecht, §9.4.1**). Monochromatic light incident normally partially reflects from (i) the lower curved surface of the lens and (ii) the top flat surface of the glass plate. Ray (ii) undergoes a phase shift of π (half-wave loss) upon reflection from the optically denser glass, while ray (i) does not.

The air-film thickness t at distance r from the point of contact (lens radius of curvature R) is:

$$t \approx \frac{r^2}{2R}$$

The conditions for dark rings (including the $\lambda/2$ phase shift) are:

$$2t = m\lambda \implies r_m^2 = m\lambda R \quad (m = 0, 1, 2, \ldots)$$

For bright rings:

$$2t = \left(m - \tfrac{1}{2}\right)\lambda \implies r_m^2 = \left(m - \tfrac{1}{2}\right)\lambda R$$

The **central spot is dark** because at the point of contact $t = 0$, the only phase difference is the π shift from reflection, giving destructive interference. The rings are **circular** due to the rotational symmetry of the geometry.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Experimental setup or image and Description	1	**2**
Conditions for constructive and destructive interference	1	

Q8

Write down the mathematical expression for the resultant irradiance in a double-slit experiment and define the symbols used. [U / CO3]

The resultant irradiance at any point on the screen in Young's double-slit experiment is (**Hecht, §9.3.1**):

$$I = 4I_0 \cos^2\left(\frac{\delta}{2}\right)$$

Symbol definitions:

- I resultant irradiance at the observation point.
- I_0 irradiance due to either slit alone.
- δ phase difference between the two waves at the observation point
  $\delta = k(r_1 - r_2) = \dfrac{2\pi}{\lambda}\, a \sin\theta \approx \dfrac{2ya\pi}{\lambda s}$
- a centre-to-centre slit separation.
- y distance of observation point from central maximum on the screen.
- s distance from slits to screen.
- λ wavelength of light.

Maxima (bright fringes): $\delta = 2m\pi$, $m = 0, \pm1, \pm2, \ldots \Rightarrow I_{\max} = 4I_0$.
Minima (dark fringes): $\delta = (2m+1)\pi \Rightarrow I_{\min} = 0$.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct formula $I = 4I_0 \cos^2(\delta/2)$	1	**2**
Correct definition of all symbols (at minimum δ, I_0, a, s)	1	

Q9

Why do two separate light sources not normally generate sustained interference fringes? [U / CO3]

For stable, observable interference fringes, the two sources must maintain a **constant phase relationship** over the observation time they must be mutually **coherent** (**Hecht, §9.2 & §7.1**). Two independent sources (two separate bulbs, two separate lasers) emit light from statistically independent atomic transitions. Each emission event lasts only $\sim 10^{-8}$ s (the coherence time τ_c), after which the phase changes randomly. The phase difference $\delta(t) = \phi_1(t) - \phi_2(t)$ therefore **fluctuates randomly with time**.

A detector integrates the intensity over a time far longer than τ_c. The time-averaged intensity is:

$$\langle I \rangle = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\langle \cos\delta(t) \rangle = I_1 + I_2$$

since $\langle \cos\delta(t) \rangle = 0$ for random phase the interference term vanishes and no fringes are seen. Sustained fringes require both beams to be derived from the **same source point** (e.g., via a single slit and two slits as in Young's experiment, or a laser).

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Problems with independent (different) sources	1	**2**
Benefits of a single source split into two sources	1	

Q10

Why is precise alignment important in the Michelson interferometer? [A / CO3]

In the Michelson interferometer, two mirrors M_1 and M_2 reflect the split beam back to a beam-splitter, where they recombine and interfere (**Hecht, §9.84.2**). Precise alignment is important because the interference pattern depends on the relative orientation of the mirrors. If the mirrors are perfectly aligned, the wavefronts match properly and produce clear, high-contrast fringes that shift uniformly with mirror movement. Any slight misalignment introduces a variation in optical path difference across the beam, causing distorted or closely spaced fringes and reducing visibility, which leads to errors in measurements.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any reasonable explanation	2	**2**

Q11

What are the conditions for secondary maxima in a multi-slit Fraunhofer diffraction pattern? [U / CO4]

The intensity distribution of N-slit Fraunhofer diffraction is (**Hecht, §10.2.3**):

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \left(\frac{\sin N\alpha}{\sin\alpha}\right)^2$$

where $\beta = \frac{kb\sin\theta}{2}$ (single-slit factor) and $\alpha = \frac{ka\sin\theta}{2}$ (multi-slit factor).$k = \dfrac{2\pi}{\lambda}$

Principal maxima occur when $\alpha = n\pi$ (n integer), i.e., $a\sin\theta_n = n\lambda$.

Secondary maxima $\sin N\beta = \pm 1, \beta = \dfrac{(2m+1)\pi}{2N}$ i.e. $a\sin\theta_m = \dfrac{(2m+1)\pi}{2N}$, The position of the mth secondary maxima near the nth principal maximum is given by

$$a\sin\theta_{(n+m)} = n\lambda + \frac{(2n+1)\pi}{2N}\lambda$$

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Intensity formula with $\sin N\alpha/\sin\alpha$ factor	1	**2**
Condition for secondary maxima	1	

Q12

What is the path difference between two consecutive half-period zones? [U / CO4]

In Fresnel's half-period zone (HPZ) construction, the wavefront beyond an aperture is divided into concentric annular zones such that the optical path from the outer edge of the m-th zone to the axial point P exceeds the path from the central point by $m\lambda/2$ (**Hecht, §10.3.1**).

The path difference between the outer edge of the m-th zone and the outer edge of the $(m+1)$-th zone is:

$$\Delta_{\text{consecutive}} = \frac{\lambda}{2}$$

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct answer: path difference $= \lambda/2$ between consecutive zones	2	**2**

Q13

What are the necessary conditions for obtaining Fraunhofer diffraction? [U / CO4]

Fraunhofer (far-field) diffraction is the limiting case of Kirchhoff diffraction in which all quadratic phase terms are negligible, leaving only a Fourier transform (**Hecht, §10.2.1**). The necessary conditions are:

1. **Plane incident wavefront:** the source must be effectively at infinity (or a collimating lens placed in front of the aperture).
2. **Observation at infinity:** the screen must be effectively at infinity (or a converging lens placed after the aperture focuses the diffracted rays at its focal plane).
3. **Fresnel number condition:**

$$N_F = \frac{a^2}{\lambda L} \ll 1$$

where a is the aperture width and L is the source-to-aperture (or aperture-to-screen) distance. When $N_F \ll 1$, the quadratic Fresnel phase terms vanish and the far-field (Fraunhofer) approximation is valid.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any two points with explanation	2	**2**

Q14

Discuss the dependence of the width of the central diffraction maximum on the wavelength of light in single-slit diffraction. [A / CO4]

In single-slit Fraunhofer diffraction the intensity is (**Hecht, §10.2.1**):

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta = \frac{\pi a \sin\theta}{\lambda}$$

The **first minima** on either side of the central maximum occur when $\beta = \pm\pi$, i.e., $a\sin\theta = \pm\lambda$:

$$\sin\theta_{\min} = \pm\frac{\lambda}{a} \implies \theta_{\min} \approx \pm\frac{\lambda}{a} \text{ (small angle)}$$

The **angular half-width** of the central maximum is $\theta_{\min} = \lambda/a$, and the **full angular width** is $2\lambda/a$.

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Intensity formula	1	**2**
First minimum condition $a\sin\theta = \pm\lambda$	1	

PART B Short Essay Questions [5 × 6 = 30 Marks]

Answer any 6 questions. Each question carries 5 marks.

Q15

Using Huygens' construction, explain the formation of ordinary and extraordinary wavefronts in a uniaxial crystal. [U / CO2]

Certain transparent crystals, known as birefringent or **doubly refracting** crystals (e.g., calcite, quartz), split an incident ray into two distinct refracted rays that travel at different speeds and in different directions. This is one of the most important phenomena in crystal optics and was first explained geometrically by Huygens in 1690 (**Hecht, §8.4**).

Uniaxial Crystal — Refractive Indices. A uniaxial crystal has a single symmetry direction called the **optic axis**. Along the optic axis the crystal is isotropic; perpendicular to it, the refractive index is different. Two refractive indices characterise the crystal:

- **Ordinary index** n_o: the same for all directions; governs the ordinary (O) ray.
- **Extraordinary index** n_e: varies with direction from n_o (along optic axis) to n_e (perpendicular); governs the extraordinary (E) ray.
- *Positive* uniaxial: $n_e > n_o$ (e.g., quartz). *Negative* uniaxial: $n_e < n_o$ (e.g., calcite).

Huygens' Secondary Wavelets. When a plane wavefront strikes the crystal surface, each point of the refracted wavefront acts as a source of **two families of secondary wavelets** that spread inside the crystal:

1. **Ordinary wavelet (O-wavelet):** Travels with speed $v_o = c/n_o$ in *all* directions. It is therefore a **sphere** of radius $v_o t$ after time t. The O-ray obeys Snell's law in the normal manner, and its electric field vibrates **perpendicular** to the plane containing the ray and the optic axis.
2. **Extraordinary wavelet (E-wavelet):** Travels with a speed that *depends on direction* — speed v_o along the optic axis and speed $v_e = c/n_e$ perpendicular to it. It is therefore a **prolate or oblate spheroid** (ellipsoid of revolution) whose axis of symmetry coincides with the optic axis. For negative uniaxial (calcite), $v_e > v_o$, so the ellipsoid lies *outside* the sphere; for positive uniaxial, $v_e < v_o$ and the ellipsoid lies *inside*. The E-ray **does not** in general obey Snell's law, and its electric field vibrates in the plane containing the ray and the optic axis.

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Introduction: birefringence, uniaxial crystal, n_o and n_e defined	3	**5**
O-wavelet: spherical, same speed $v_o = c/n_o$ in all directions	1	
E-wavelet: ellipsoidal, direction-dependent speed; inside/outside sphere explained	1	

Q16

Explain polarisation by scattering. [K / CO2]

When light passes through a medium containing small particles (gas molecules, colloidal particles, aerosols), it is scattered in various directions. The scattered light is **partially or completely polarised** depending on the angle of observation, even when the incident light is completely unpolarised. This phenomenon is known as **polarisation by scattering**(**Hecht, §8.5.1**).

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
A comprehensive explanation of polarisation by scattering	5	**5**

Q17

Explain phasor representation of harmonic waves. Show how two sinusoidal waves of the same frequency but different phases can be added using a phasor diagram. [U / CO3]

A **phasor** is a rotating vector in a two-dimensional complex plane used to represent a harmonic (sinusoidal) wave. It transforms the algebraic addition of trigonometric functions into a simpler geometric vector (polygon) construction, which is especially powerful for multi-beam interference calculations (**Hecht, §2.6**).

Phasor Representation of a Single Harmonic Wave.
A wave $E(t) = E_0 \cos(\omega t + \phi)$ is represented by a phasor a vector of **Length** $= E_0$ (amplitude of the wave), **Angle** $= \omega t + \phi$ measured from the positive real axis.

As time progresses, the phasor rotates counterclockwise at angular frequency ω. The **instantaneous value of the wave** is the projection of the phasor onto the real axis:

$$E(t) = \mathrm{Re}\left[E_0\, e^{i(\omega t+\phi)}\right] = E_0 \cos(\omega t + \phi)$$

Addition of Two Waves by Phasor Diagram.
Consider two waves of the same angular frequency ω but different amplitudes and phases:

$$E_1 = E_{01}\cos(\omega t + \phi_1), \qquad E_2 = E_{02}\cos(\omega t + \phi_2)$$

Procedure:

1. Draw Phasor 1 as an arrow of length E_{01} at angle ϕ_1 from the real axis.
2. Place the tail of Phasor 2 (length E_{02}, angle ϕ_2) at the tip of Phasor 1 (*tip-to-tail vector addition*).
3. The **resultant phasor** runs from the origin to the tip of Phasor 2.
4. Its length gives the **resultant amplitude** E_R and its angle gives the **resultant phase** ϕ_R.

Analytical Results from Phasor Addition (parallelogram law):

$$E_R^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos(\phi_2 - \phi_1)$$

$$\tan\phi_R = \frac{E_{01}\sin\phi_1 + E_{02}\sin\phi_2}{E_{01}\cos\phi_1 + E_{02}\cos\phi_2}$$

The resultant wave is $E_R\cos(\omega t + \phi_R)$.

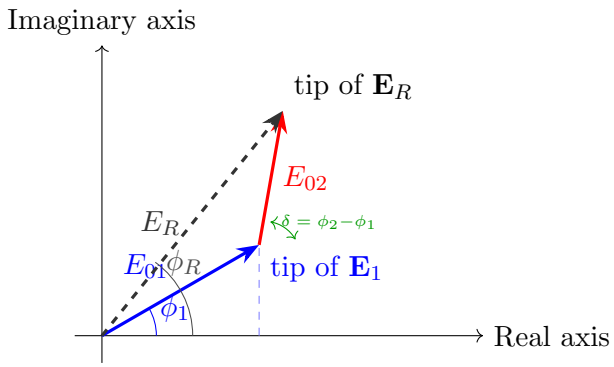


Figure 1: Phasor addition of two harmonic waves. Phasor $\mathbf{E}_1$ (blue) at angle ϕ_1, Phasor $\mathbf{E}_2$ (red) added tip-to-tail, resultant $\mathbf{E}_R$ (dashed) gives amplitude E_R and phase ϕ_R.

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Phasor definition: rotating vector, length = amplitude, angle = phase	2.5	**5**
Tip-to-tail addition procedure described clearly or phasor diagram drawn	2.5	

Q18

A wire is placed between two glass plates forming an air wedge. Derive an expression for calculating its thickness. [U / CO3]

An air wedge is one of the simplest and most elegant thin-film interferometers. By counting the interference fringes formed in the wedge-shaped air gap, the thickness of an extremely thin object - such as a wire, hair, or thin foil - can be measured with sub-micrometre precision (**Hecht, §9.4**).

Experimental Setup. Two flat glass plates are placed in contact at one end (the *apex* of the wedge) and separated at the other end by the wire of unknown diameter/thickness d. This creates a wedge-shaped air film of gradually increasing thickness. Monochromatic light of wavelength λ is incident normally on the top plate.

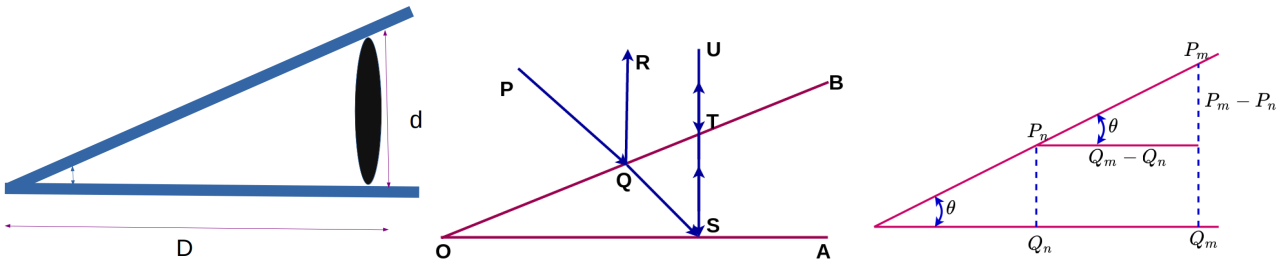


Figure 2: air wedge.

Derivation. Let:

- θ = wedge angle (very small, $\theta \ll 1$).
- D = total length of the wedge (= distance from apex to wire).
- d = diameter of the wire (= t at $x = L$).

Thickness of the air-film for a bright fringe at P_n	$2P_nQ_n$	$= (2n+1)\,\dfrac{\lambda}{2}$
Thickness of the air-film for a bright fringe at P_m	$2P_mQ_m$	$= (2m+1)\,\dfrac{\lambda}{2}$
	$P_mQ_m - P_nQ_n$	$= (m-n)\,\dfrac{\lambda}{2}$

$$
\begin{aligned}
\theta &= \frac{P_mQ_m - P_nQ_n}{Q_m - Q_n} = \frac{(m-n)\frac{\lambda}{2}}{x} \\
x &= \frac{(m-n)\frac{\lambda}{2}}{\theta} \\
\beta &= \frac{x}{m-n} = \frac{\lambda}{2\theta} \\
\theta &= \frac{\lambda}{2\beta} = \frac{d}{D} \\
d &= \frac{\lambda\, D}{2\beta} \qquad (1)
\end{aligned}
$$

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Setup described: wedge geometry,figure	2	**5**
Condition for dark/bright fringes	1	
Derivation	2	

Q19

Discuss the interference of reflected light from a thin film. Deduce the conditions for constructive and destructive interference. [A / CO3]

The vivid iridescent colours of soap bubbles, oil films on water, and anti-reflection lens coatings all arise from **thin-film interference** - the superposition of light waves reflected from the two surfaces of a thin, transparent film (**Hecht, §9.4**). This phenomenon is one of the most beautiful consequences of wave optics and has profound practical applications in optical coatings.

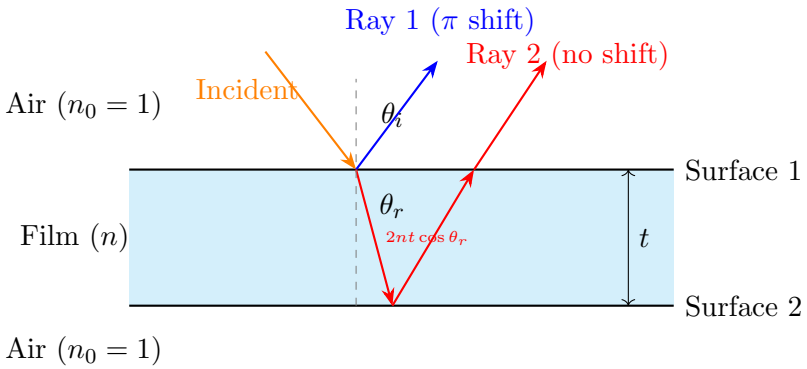


Figure 3: Thin-film interference. Ray 1 reflects from the top surface with a π phase shift; Ray 2 traverses the film twice with optical path $2nt\cos\theta_r$ and no phase shift at the bottom surface. Their superposition produces constructive or destructive interference.

Setup. A thin transparent film of thickness t and refractive index n ($n > n_{\text{air}} = 1$) is surrounded by air on both sides. Light of wavelength λ (in vacuum) is incident at angle θ_i in air. By Snell's law, the refraction angle inside the film is θ_r where $\sin\theta_i = n\sin\theta_r$.

Two Reflected Rays.

- **Ray 1** reflects from the *top surface* (air $\rightarrow$ film, going from lower n to higher n): undergoes a **phase shift of** π (half-wave loss), equivalent to an extra optical path of $\lambda/2$.
- **Ray 2** refracts into the film, reflects from the *bottom surface* (film $\rightarrow$ air, going from higher n to lower n): **no phase shift** on reflection. Ray 2 then refracts back out through the top surface.

Optical Path Difference. The path traversed by Ray 2 inside the film (going down and back up) is $2t/\cos\theta_r$ (geometric path in the film), but must be multiplied by the film's refractive index n to get the optical path. Correcting also for the lateral shift, the net optical path difference between Rays 1 and 2 is (**Hecht, §9.4**):

$$\Delta_{\text{optical}} = 2nt\cos\theta_r$$

Including the π phase shift of Ray 1 (equivalent to $+\lambda/2$), the **effective path difference** is:

$$\Delta_{\text{eff}} = 2nt\cos\theta_r + \tfrac{\lambda}{2}$$

Conditions for Constructive Interference (Bright reflection): Constructive interference requires the effective path difference to equal an integer number of wavelengths:

$$2nt\cos\theta_r + \tfrac{\lambda}{2} = m\lambda, \quad m = 1, 2, 3, \ldots$$

$$\therefore \quad \boxed{2nt\cos\theta_r = \left(m - \tfrac{1}{2}\right)\lambda = \frac{(2m-1)\lambda}{2}}$$

Conditions for Destructive Interference (Dark / no reflection): Destructive interference requires the effective path difference to equal a half-integer multiple of λ:

$$2nt\cos\theta_r + \tfrac{\lambda}{2} = \left(m + \tfrac{1}{2}\right)\lambda$$

$$\therefore \quad \boxed{2nt\cos\theta_r = m\lambda}, \quad m = 0, 1, 2, \ldots$$

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Setup: film geometry, two reflected rays identified	1	
Phase analysis: π shift at top surface; no shift at bottom surface	1	
Optical path difference $\Delta = 2nt\cos\theta_r$ derived	1	**5**
condition for constructive($2nt\cos\theta_r = (2m-1)\lambda/2$) or Destructive ($2nt\cos\theta_r = m\lambda$)	2	

Q20

Explain the Huygens–Fresnel principle and describe how it accounts for the propagation of light waves. [U/CO4]

The Huygens–Fresnel Principle is one of the most fundamental concepts in wave optics. It provides a complete physical explanation for the propagation of light waves and the phenomenon of diffraction, which is the bending of light around obstacles and apertures.(**Hecht, §10.1**)

Original Huygens's Principle

Christian Huygens proposed that every point on a wavefront at a given instant acts as a source of secondary spherical wavelets. The new wavefront at any later instant is simply the envelope of all these secondary wavelets. While this construction successfully predicted the general direction of wave propagation, it had a serious limitation. It made no reference to wavelength and ignored the mutual interaction of the secondary wavelets. As a result, it could not explain why light forms sharp geometric shadows while sound bends easily around large obstacles. The principle was geometrically useful but physically incomplete.

Fresnel's Contribution

Fresnel overcame this limitation by introducing the concept of interference into Huygens's construction. He proposed that the secondary wavelets emitted from each point on the wavefront do not merely form an envelope but superpose with one another. The resultant optical field at any point in space is determined by the combined effect of all these wavelets, taking into account both their amplitudes and their relative phases. This single addition of interference transformed the principle from a geometrical approximation into a powerful physical theory capable of predicting the detailed structure of diffraction patterns.

Huygens–Fresnel Principle The Huygens–Fresnel Principle states that every unobstructed point on a wavefront, at a given instant, serves as a source of secondary spherical wavelets having the same frequency as the primary wave. The amplitude of the optical field at any point beyond is the superposition of all these wavelets, considering their amplitudes and relative phases.

How it Accounts for Propagation of Light Waves

The principle explains light propagation through the mechanism of interference among secondary wavelets as follows:

In the **forward direction**, the secondary wavelets from all points on the wavefront travel nearly equal path lengths and arrive approximately in phase. They therefore interfere constructively, producing a strong resultant field and sustaining the forward propagation of the wave.

In **oblique directions**, the path lengths from different points on the wavefront to the observation point differ significantly. The wavelets arrive with varying phases and interfere destructively, suppressing the optical field in those directions.

When the **wavelength is large** compared to the aperture, the wavelets interfere constructively over a wide angular range, causing the wave to spread broadly into the shadow region. This explains the strong diffraction of sound around everyday objects.

When the **wavelength is small** compared to the aperture, constructive interference is confined to a narrow forward cone, and a well-defined geometric shadow is formed. This explains why light appears to travel in straight lines under ordinary conditions.

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Huygens'/Huygens–Fresnel principle	2	**5**
How it Accounts for Propagation of Light Waves	3	

Q21

Sketch and compare the fringe patterns obtained in Young's double-slit experiment and in Fraunhofer double-slit diffraction. [A/CO4]

Young's double-slit experiment treats the slits as infinitesimally narrow (ideal) point sources and considers only the **interference** between the two beams. Fraunhofer double-slit diffraction is the more physically accurate treatment that accounts for the **finite width** b of each slit, leading to a pattern that is the product of a single-slit diffraction envelope and a two-slit interference pattern (**Hecht, §9.3.1 and §10.2.2**).

A. Young's Double-Slit — Intensity Formula.

The two infinitely narrow slits separated by a act as coherent line sources. The irradiance at angle θ is:

$$I_{\text{Young}}(\theta) = 4I_0\cos^2(\alpha) = 4I_0\cos^2\left(\tfrac{ya\pi}{s\lambda}\right), \qquad \alpha = \tfrac{\pi a\sin\theta}{\lambda} = \tfrac{ya\pi}{s\lambda}$$

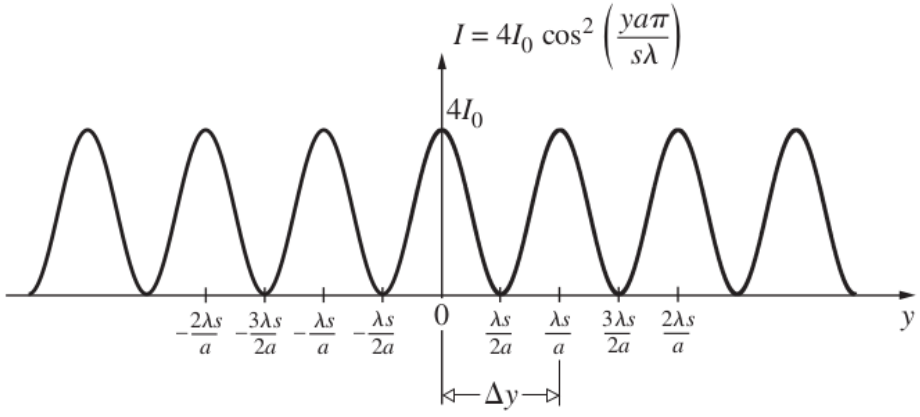


Figure 4: Intensity patterns for Young's double-slit

B. Fraunhofer Double-Slit — Intensity Formula.

Each slit of width b produces its own single-slit diffraction pattern. The combined intensity is the **product** of the single-slit diffraction envelope and the two-slit interference pattern:

$$I(\theta) = 4I_0\left(\frac{\sin\beta}{\beta}\right)^2\cos^2\alpha$$

$$\alpha = \frac{\pi a \sin\theta}{\lambda} = \frac{ya\pi}{s\lambda} \quad \text{(double-slit interference factor)}$$

$$\beta = \frac{\pi b \sin\theta}{\lambda} = \frac{yb\pi}{s\lambda} \quad \text{(single-slit diffraction factor)}$$

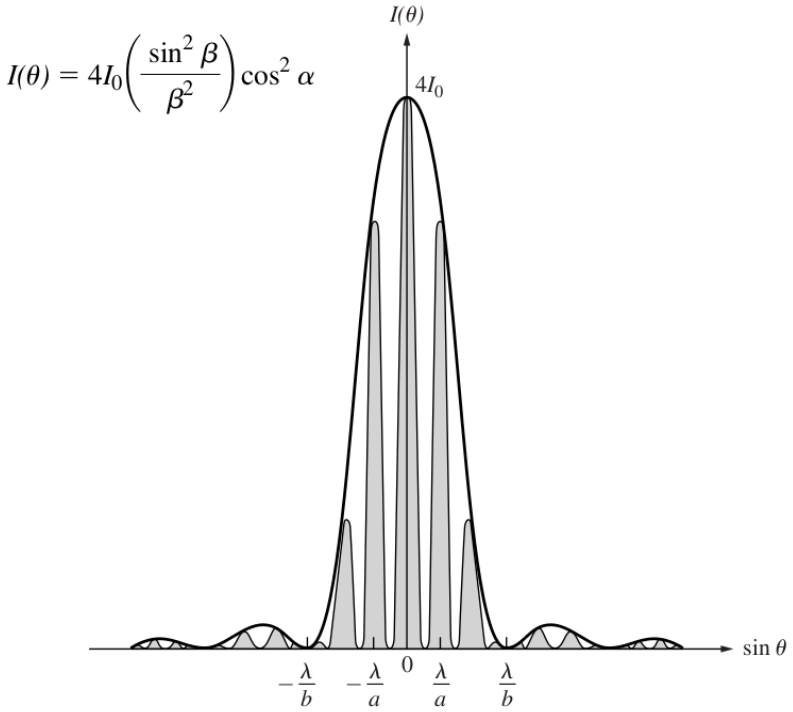


Figure 5: Intensity patterns for Fraunhofer double-slit

Comparison Table:

Feature	Young's (ideal)	Fraunhofer double-slit
Slit width	Infinitely narrow (ideal point source)	Finite width b
Nature of pattern	Pure interference	Interference modulated by diffraction
Intensity formula	$4I_0\cos^2\alpha$	$4I_0(\sin\beta/\beta)^2\cos^2\alpha$
Fringe spacing	λ/d	λ/d (same)
Fringe heights	All equal ($4I_0$)	Modulated by sinc^2 envelope
Central max	Same as all others	Brightest, others diminish outward

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Young's formula $I = 4I_0\cos^2(\alpha)$; all fringes equal; Graphs/sketches	2	**5**
Fraunhofer formula $I = 4I_0(\sin\beta/\beta)^2\cos^2\alpha$; modulation explained;Graphs/sketches	2	
Comparison table	1	

Q22

Describe the construction of a zone plate. [U / CO4]

A **zone plate** is an optical device that focuses light by the principle of diffraction, unlike a conventional lens which focuses light by refraction. It is constructed based on the concept of Fresnel half-period zones and works by blocking out alternate zones so that only constructive interference contributions reach the focal point, resulting in a bright focused spot (**Hecht, §10.3.5**).

Concept of Fresnel Half-Period Zones Consider a plane wavefront advancing toward a point of observation P. The wavefront can be divided into a series of concentric circular zones called Fresnel half-period zones, such that the distance from the outer edge of each successive zone to P differs by $\lambda/2$ from the previous one. If $r_1,\ r_2,\ r_3,\ \ldots$ are the radii of successive zones, then the path difference between consecutive zones is $\lambda/2$. The contributions from alternate zones are in opposite phase, meaning odd zones and even zones tend to cancel each other out. The resultant amplitude at P due to the entire unobstructed wavefront is therefore only about half the contribution of the first zone alone.

Radius of the nth Fresnel zone:

$$R_m = \sqrt{mr_0\lambda}$$

where:
m is zone number, λ wavelength of light, r_0 distance from the wavefront to the point of observation P

A zone plate is **constructed** by first drawing a highly magnified diagram on paper with concentric circles having radii proportional to $\sqrt{1},\ \sqrt{2},\ \sqrt{2},\cdots$ representing the boundaries of successive Fresnel half-period zones. Alternate zones are then shaded black while the remaining zones are left transparent, producing a pattern of alternating opaque and transparent concentric rings. This drawing is then photographed and reduced to a very small size using photographic techniques, ensuring that the zone radii are of the order of millimetres or less, appropriate for use with visible light. The developed photographic film or glass plate with this pattern of alternating opaque and transparent rings constitutes the finished zone plate, which can then be mounted and used in place of a converging lens.

Focal Length The zone plate acts as a converging lens with focal length: $f_1 = \dfrac{R_m^2}{m\lambda}$ It produces multiple focal points at $f,\ f/3,\ f/5,\ \cdots$ unlike a simple lens which has only one focal point.

A **zone plate** is constructed by blocking alternate Fresnel zones on a photographic plate. It focuses light by diffraction and serves as a diffractive equivalent of a converging lens, with the unique property of possessing multiple focal points.

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Fresnel half-period zones: definition, radii $R_m = \sqrt{mr_0\lambda}$, equal areas	1	**5**
Why adjacent zones cancel in free propagation (π phase difference)	1	
Construction: blocking alternate zones step-by-step; photographic reduction	3	

Note: Examiners may award marks proportionally for partially correct answers. Credit should be given for relevant diagrams, intermediate steps, correct identification of formulas, and clear physical interpretation even if the final result contains minor errors.

End of Scheme of Valuation — All answers prepared with reference to Optics *by Eugene Hecht (5th Ed.)*

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) Regular Examination — March 2026

Fourth Semester

Course Code MG4DSCPHY200

Question Paper Code 25901435

WAVE OPTICS

Scheme for Valuation with Model Answers

Reference: Optics *by Eugene Hecht & A.R.Ganesan (5th Edition)*

Mark Distribution Summary

Maximum Marks: **50** Duration: **1 hour 30 minutes**

Part	Question Type	No. to Attempt	Marks Each	Total
Part A	Short Answer Questions	Any 10 out of 12	2	**20**
Part B	Short Essay	Any 6 out of 8	5	**30**
			TOTAL MARKS	**50**

PART A Short Answer Questions [2 × 10 = 20 Marks]

Answer any 10 questions. Each question carries 2 marks.

Q1

What is the relation between wave vector and wavelength? [U / CO1]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct relation $\mathbf{k} = 2\pi/\lambda$ with units / symbols	2	**2**

Q2

Formulate the wave equation in three dimensions. [K / CO1]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct 3D wave equation with Laplacian notation or in Cartesian expansion	1	**2**
v is the phase velocity or speed of the wave in the medium	1	

Q3

Mention any four quantities that describe a wave. [K / CO1]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any four correct quantities named	1	**2**
Brief description or equation for each quantity	1	

Q4

In a polarising device, what is referred to as the extinction axis? [U / CO2]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Extinction axis = direction perpendicular to transmission axis; absorbed component	2	**2**

Q5

Why is quartz said to be optically active? [U / CO2]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Definitions for optically active crystal	1	**2**
Rotation formula and optical activity explained	1	

Q6

What phase difference corresponds to a path difference of $\lambda/2$? [U / CO3]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct formula $\delta = (2\pi/\lambda)\Delta$	2	**2**

Q7

Explain the formation of Newton's rings. [K / CO3]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Experimental setup or image and Description	1	**2**
Conditions for constructive and destructive interference	1	

Q8

Write down the mathematical expression for the resultant irradiance in a double-slit experiment and define the symbols used. [U / CO3]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct formula $I = 4I_0 \cos^2(\delta/2)$	1	**2**
Correct definition of all symbols (at minimum δ, I_0, a, s)	1	

Q9

Why do two separate light sources not normally generate sustained interference fringes? [U / CO3]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Problems with independent (different) sources	1	**2**
Benefits of a single source split into two sources	1	

Q10

Why is precise alignment important in the Michelson interferometer? [A / CO3]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any reasonable explanation	2	**2**

Q11

What are the conditions for secondary maxima in a multi-slit Fraunhofer diffraction pattern? [U / CO4]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Intensity formula with $\sin N\alpha / \sin\alpha$ factor	1	**2**
Condition for secondary maxima	1	

Q12

What is the path difference between two consecutive half-period zones? [U / CO4]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Correct answer: path difference $= \lambda/2$ between consecutive zones	2	**2**

Q13

What are the necessary conditions for obtaining Fraunhofer diffraction? [U / CO4]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Any two points with explanation	2	**2**

Q14

Discuss the dependence of the width of the central diffraction maximum on the wavelength of light in single-slit diffraction. [A / CO4]

Valuation Scheme — 2 Marks

Component / Answer Element	Marks	Total
Intensity formula	1	**2**
First minimum condition $a\sin\theta = \pm\lambda$	1	

PART B Short Essay Questions [5 × 6 = 30 Marks]

Answer any 6 questions. Each question carries 5 marks.

Q15

Using Huygens' construction, explain the formation of ordinary and extraordinary wavefronts in a uniaxial crystal. [U / CO2]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Introduction: birefringence, uniaxial crystal, n_o and n_e defined	3	**5**
O-wavelet: spherical, same speed $v_o = c/n_o$ in all directions	1	
E-wavelet: ellipsoidal, direction-dependent speed; inside/outside sphere explained	1	

Q16

Explain polarisation by scattering. [K / CO2]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
A comprehensive explanation of polarisation by scattering	5	**5**

Q17

Explain phasor representation of harmonic waves. Show how two sinusoidal waves of the same frequency but different phases can be added using a phasor diagram. [U / CO3]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Phasor definition: rotating vector, length = amplitude, angle = phase	2.5	**5**
Tip-to-tail addition procedure described clearly or phasor diagram drawn	2.5	

Q18

A wire is placed between two glass plates forming an air wedge. Derive an expression for calculating its thickness. [U / CO3]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Setup described: wedge geometry,figure	2	**5**
Condition for dark/bright fringes	1	
Derivation	2	

Q19

Discuss the interference of reflected light from a thin film. Deduce the conditions for constructive and destructive interference. [A / CO3]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Setup: film geometry, two reflected rays identified	1	5
Phase analysis: π shift at top surface; no shift at bottom surface	1	
Optical path difference $\Delta = 2nt\cos\theta_r$ derived	1	
condition for constructive($2nt\cos\theta_r = (2m-1)\lambda/2$) or Destructive ($2nt\cos\theta_r = m\lambda$)	2	

Q20

Explain the Huygens–Fresnel principle and describe how it accounts for the propagation of light waves. [U/CO4]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Huygens'/Huygens–Fresnel principle	2	5
How it Accounts for Propagation of Light Waves	3	

Q21

Sketch and compare the fringe patterns obtained in Young's double-slit experiment and in Fraunhofer double-slit diffraction. [A/CO4]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Young's formula $I = 4I_0\cos^2(\alpha)$; all fringes equal; Graphs/sketches	2	5
Fraunhofer formula $I = 4I_0(\sin\beta/\beta)^2\cos^2\alpha$; modulation explained;Graphs/sketches	2	
Comparison table	1	

Q22

Describe the construction of a zone plate. [U / CO4]

Valuation Scheme — 5 Marks

Component / Answer Element	Marks	Total
Fresnel half-period zones: definition, radii $R_m = \sqrt{mr_0\lambda}$, equal areas	1	5
Why adjacent zones cancel in free propagation (π phase difference)	1	
Construction: blocking alternate zones step-by-step; photographic reduction	3	

Note: Examiners may award marks proportionally for partially correct answers. Credit should be given for relevant diagrams, intermediate steps, correct identification of formulas, and clear physical interpretation even if the final result contains minor errors.